

Multiple Choice (10 marks)

1. According to the Solar Nebular theory what are asteroids and comets?
 - (a) They are the shattered remains of collisions between planets.
 - (b) They are chunks of rock or ice that condensed long after the planets and moons had formed.
 - (c) They are chunks of rock or ice that were expelled from planets by volcanoes.
 - (d) They are leftover planetesimals that never accreted into planets.
2. Approximately how far away is the Andromeda Galaxy?
 - (a) 1.7 million light years
 - (b) 2.1 million light years
 - (c) 2.5 million light years
 - (d) 3.2 million light years
3. Moons cause/contribute to which of the following?
 - (a) stability of particles within rings.
 - (b) gravitational effects at ring edges as the moons pass by.
 - (c) gaps between rings.
 - (d) Moons contribute to all of the above.
4. What is not true of Jupiter's magnetic field?
 - (a) it is about 20000 times stronger than Earth's magnetic field
 - (b) it traps charged particles from Io's volcanoes in a "plasma torus" around the planet
 - (c) it causes spectacular auroral displays at Jupiter's north and south poles
 - (d) it switches polarity every 11 years
5. How do astronomers think Jupiter generates its internal heat?
 - (a) nuclear fusion in the core
 - (b) by contracting changing gravitational potential energy into thermal energy
 - (c) internal friction due to its high rotation rate
 - (d) chemical processes

True / False (10 marks)

Answer True or False

6. True or False: The nebular theory does not predict an equal number of terrestrial and jovian planets in the solar system.
7. True or False: Volcanic heating cannot account for Mercury's lack of a permanent atmosphere due to its insufficient volcanic activity.
8. True or False: A solar day on Planet X lasts approximately 100 Earth days, equal to its sidereal day length.
9. True or False: Type-Ia supernovae typically produce high amounts of X-rays, making them prominent sources of X-ray emissions in the universe.
10. True or False: The azimuth of the north celestial pole (NCP) indicates your geographic latitude in the sky.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the unique characteristics of asteroids that differentiate them from background stars in sky surveys, focusing on their motion and implications for observation techniques.
12. Discuss the formation of "hot Jupiters" around other stars, analyzing their origin beyond the frost line and the mechanisms that lead to their inward migration.

End of Paper